Advanced Macroeconomic Analysis I

Winter 2026/27

Time and place

Wednesdays, 10-12 am & Thursdays, 10-12 am

First part (starting October 14th): Wednesdays FU Berlin, Garystr. 21, HS 106 & Thursdays, tba

Second part (starting December 9th) : Elinor-Ostrom room, DIW Berlin

Instructors

Prof. Dr. Britta Gehrke, Prof. Dr. Alexander Kriwoluzky, Dr. Nicolas Syrichas

Alina Ermolaeva, Jo-Ya Kung, Dr. Florian von Muschwitz

Summary

The course "Advanced Macroeconomic Analysis I" is the first semester of the full-year 1st year PhD sequence in macroeconomics. It serves as an introduction to methods and models of macroeconomics and to train students in basic dynamic concepts and tools used by researchers in macroeconomics, microeconomics, and econometrics.

This course is open to BSoE students, qualified master's students, or others with the instructor's permission. The course presumes familiarity with macroeconomics. The course is at an advanced level and we will move rapidly through difficult material. It is not a course for those encountering macroeconomics for the first time.

Evaluation Mid-term and final exam (120 min)

Textbooks

- M. Azzimonti, P. Krusell, A. McKay, and T. Mukoyama et al. (2026). Macroeconomics. A comprehensive textbook for first-year Ph.D. courses in macroeconomics. https://phdmacrobook.org/
- Gali, J. (2015). Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications, Princeton University Press.
- Ljungqvist, L., and T. Sargent (2018). Recursive Macroeconomic Theory 4nd ed. Cambridge, USA: MIT Press.

Part 1: Introduction to Modern Macroeconomics (Britta Gehrke and Nicolas Syrichas, week 1–8)

Description

This part of the course presents standard frameworks used in modern macroeconomics. It focuses on dynamic optimization, dynamic programming and the solution of dynamic models without and with uncertainty in the context of the neoclassical growth model and with labor market frictions.

Overview (preliminary)

1. Week 1: Introduction and framework
    - Readings: AKMM Chapters 1+2, Lucas (1976), Sims (1980) Nakamura and Steinsson (2018)
2. Week 2: Consumption and saving: Dynamic optimization, general equilibrium, recursive problems, value function
    - Readings: AKMM Chapters 4+5, LS Chapters 1+3
3. Week 3+4: The neoclassical growth model: Solving dynamic models, steady state, phase diagram, dynamic effects of shocks, planner vs decentralized equilibrium, solving linearized models, value function iteration
    - Readings: AKMM Chapters 4-6 + 10, Campbell (1994), Blanchard and Kahn (1980a)
4. Week 4+5: Frictional labor markets: Uncertainty, Markov processes, expectations, impulse response functions, aggregate vs idiosyncratic risk
    - Readings: AKMM Chapters 7, 10+20, Pissarides, C. (2000). *Equilibrium Unemployment Theory*, 2nd ed. MIT Press.
5. Week 6: Overlapping generations: equilibrium, efficiency, and policy (Nicolas Syrichas)

    Two-period lives; exchange and production economies; competitive equilibrium and capital accumulation; the golden rule and dynamic inefficiency; pay-as-you-go pensions and public debt.

    - Required readings: AKMM Chapters 5–6 (OLG sections); Krueger, Sections 8.1 and 8.3 (selected material).

- Further reading: Krueger, Section 8.2 (Ricardian equivalence); AKMM Chapter 15 (OLG and intergenerational policy sections).
- Krueger, D. (2012). *Macroeconomic Theory.* Lecture notes, University of Pennsylvania, January 26 version.

6. Week 7: Continuous-time macroeconomics I: optimization and growth (Nicolas Syrichas)

   From discrete to continuous time; discounting and flow budget constraints; the current-value Hamiltonian, dynamic programming and the Hamilton–Jacobi–Bellman equation.

   - Required readings: AKMM Chapter 9 (deterministic methods and growth); Krueger, Section 9.3.
   - Further readings: Wälde, Sections 3.2–3.4; Mukoyama, *Relationship between Maximum Principle and Dynamic Programming.*
   - Krueger, D. (2012). *Macroeconomic Theory.* Lecture notes, University of Pennsylvania, January 26 version.
   - Mukoyama, T. Selected supplementary notes: *DMP Model in Continuous Time* and *Relationship between Maximum Principle and Dynamic Programming.*

7. Week 8: Continuous-time macroeconomics II: uncertainty and labor markets (Nicolas Syrichas)

   Poisson processes and transition rates; Bellman equations with jumps; worker and firm values in continuous time; unemployment flows and the Beveridge curve; vacancy creation and wage bargaining.

   - Required readings: AKMM Chapter 9 (Poisson uncertainty); Mukoyama, *DMP Model in Continuous Time.*
   - Further readings: Wälde, Sections 5.3, 5.6.1, and 8.2–8.4; AKMM Chapter 20 (review of the DMP model).
   - Mukoyama, T. Selected supplementary notes: *DMP Model in Continuous Time* and *Relationship between Maximum Principle and Dynamic Programming.*
   - Wälde, K. (2020/21). *Advanced Macroeconomic Theory 1 (Part 2).* Lecture notes, GSEFM, Johannes Gutenberg University Mainz.

Part 2: Monetary & fiscal policy in a New-Keynesian model (Alexander Kriwoluzky, week 9–16)

Overview

This part of the course is built around the New Keynesian (NK) model and the question of who controls inflation. We open with a historical lecture that poses the questions of the course, then build the model step by step: the classical monetary model, the NK model with sticky prices, and the design of monetary policy. We then add fiscal policy, first with a representative household and then with two types of households (TANK). The final two lectures investigate the consequences of the interaction between monetary and fiscal policy. From the second week on, each lecture is followed by a tutorial that implements the corresponding model in `Dynare`.

Introduction

Week 1 (9 December): A small sample of macro-history

- Main readings: Galí (2015) (Ch. 1); Friedman and Schwartz (1963) (Ch. 7); Nakamura and Steinsson (2018)

Monetary policy

Week 2 (16 December): The classical monetary model

- Main readings: Galí (2015) (Ch. 2); Walsh (2017) (Ch. 2); Woodford (2003) (Ch. 2)

Tutorial: The classical model and introduction to `Dynare`

Week 3 (6 January): The basic New Keynesian model

- Main readings: Galí (2015) (Ch. 3); Clarida et al. (1999); Christiano et al. (1999)

Tutorial: The three-equation NK model in `Dynare`

Week 4 (13 January): Monetary policy design — determinacy and optimal policy

- Readings: Galí (2015) (Ch. 4); Blanchard and Kahn (1980b); Woodford (2003) (Ch. 4)

Tutorial: Determinacy and simple rules in `Dynare`

Fiscal policy

Week 5 (20 January): Fiscal policy in the New Keynesian model

- Main readings: Woodford (2011); Christiano et al. (2011); Ramey (2019)

Tutorial: Fiscal multipliers in `Dynare`

Week 6 (27 January): Heterogeneity — the two-agent NK model (TANK) and the Keynesian cross

- Main readings: Bilbiie (2021); Bilbiie (2025); Kaplan et al. (2018)

Tutorial: RANK versus TANK in `Dynare`

Monetary–fiscal interaction

Week 7 (3 February): Why money has value

- Main readings: Sargent and Wallace (1981); Sargent (1982); Cochrane (2023) (Ch. 1–2)

Tutorial: The government budget constraint in `Dynare`

Week 8 (10 February): Monetary–fiscal interaction and the fiscal theory of the price level

- Main readings: Leeper (1991); Kliem et al. (2016); Ettmeier and Kriwoluzky (2024)

Tutorial: Exam preparation

Readings

Bilbiie, F. O. (2021). The new keynesian cross. *Journal of Monetary Economics*, 122:38–55.

Bilbiie, F. O. (2025). Monetary policy and heterogeneity: An analytical framework. *Review of Economic Studies*, 92(4):2398–2436.

Blanchard, O. J. and Kahn, C. M. (1980a). The solution of linear difference models under rational expectations. *Econometrica: Journal of the Econometric Society*, pages 1305–1311.

Blanchard, O. J. and Kahn, C. M. (1980b). The solution of linear difference models under rational expectations. *Econometrica*, 48(5):1305–1311.

Campbell, J. Y. (1994). Inspecting the mechanism: An analytical approach to the stochastic growth model. *Journal of Monetary Economics*, 33(3):463–506.

Christiano, L., Eichenbaum, M., and Rebelo, S. (2011). When is the government spending multiplier large? *Journal of Political Economy*, 119(1):78–121.

Christiano, L. J., Eichenbaum, M., and Evans, C. L. (1999). Monetary policy shocks: What have we learned and to what end? In Taylor, J. B. and Woodford, M., editors, *Handbook of Macroeconomics*, volume 1, chapter 2, pages 65–148. Elsevier, Amsterdam.

Clarida, R., Galí, J., and Gertler, M. (1999). The science of monetary policy: A new keynesian perspective. *Journal of Economic Literature*, 37(4):1661–1707.

Cochrane, J. H. (2023). *The Fiscal Theory of the Price Level*. Princeton University Press, Princeton, NJ.

Ettmeier, S. and Kriwoluzky, A. (2024). Active or passive? Revisiting the role of fiscal policy during high inflation. *European Economic Review*, 170.

Friedman, M. and Schwartz, A. J. (1963). *A Monetary History of the United States, 1867–1960*. Princeton University Press, Princeton, NJ.

Galí, J. (2015). *Monetary Policy, Inflation, and the Business Cycle: An Introduction to the New Keynesian Framework and Its Applications*. Princeton University Press, Princeton, NJ, 2 edition.

Kaplan, G., Moll, B., and Violante, G. L. (2018). Monetary policy according to hank. *American Economic Review*, 108(3):697–743.

Kliem, M., Kriwoluzky, A., and Sarferaz, S. (2016). Monetary-fiscal policy interaction and fiscal inflation: A tale of three countries. *European Economic Review*.

Leeper, E. M. (1991). Equilibria under 'active' and 'passive' monetary and fiscal policies. *Journal of Monetary Economics*, 27(1):129–147.

Lucas, R. E. (1976). Econometric policy evaluation: A critique. *Carnegie-Rochester Conference Series on Public Policy*, 1:19–46.

Nakamura, E. and Steinsson, J. (2018). Identification in macroeconomics. *Journal of Economic Perspectives*, 32(3):59–86.

Ramey, V. A. (2019). Ten years after the financial crisis: What have we learned from the renaissance in fiscal research? *Journal of Economic Perspectives*, 33(2):89–114.

Sargent, T. J. (1982). The ends of four big inflations. In Hall, R. E., editor, *Inflation: Causes and Effects*, pages 41–97. University of Chicago Press, Chicago.

Sargent, T. J. and Wallace, N. (1981). Some unpleasant monetarist arithmetic. *Federal Reserve Bank of Minneapolis Quarterly Review*, 5(3):1–17.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica: Journal of the Econometric Society*, pages 1–48.

Walsh, C. E. (2017). *Monetary Theory and Policy*. MIT Press, Cambridge, MA, 4 edition.

Woodford, M. (2003). *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press, Princeton, NJ.

Woodford, M. (2011). Simple analytics of the government expenditure multiplier. *American Economic Journal: Macroeconomics*, 3(1):1–35.